

Intelligent NER & Semantic Search System

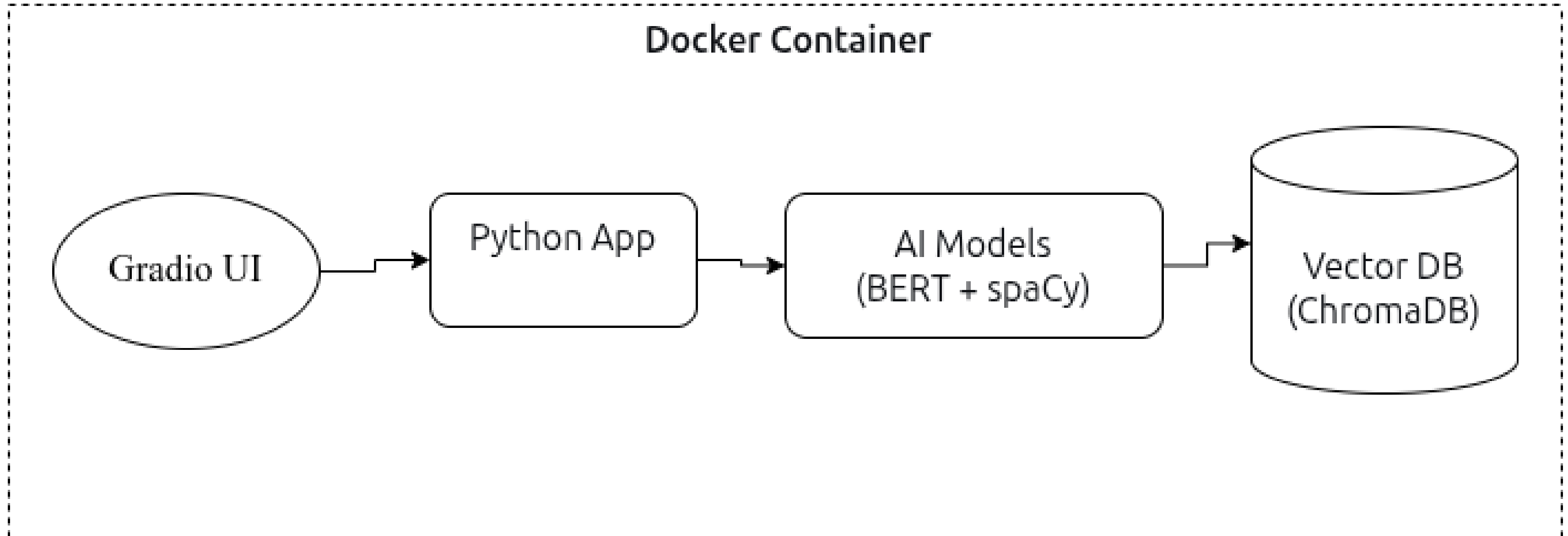
Course: Professional Training (Intermediate) | Instructor: Ahmed Zahir

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Problem & Motivation

- **Problem:** Massive amounts of unstructured text (PDFs, articles) are hard to search and analyze.
- **Solution:** An automated pipeline that identifies Who, What, and Where and allows searching by meaning, not just keywords.

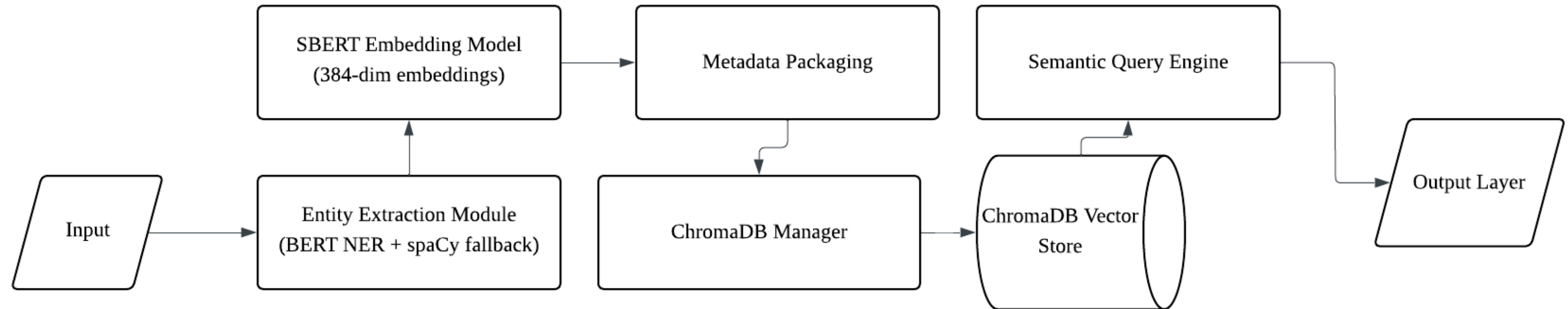
System Architecture



The Intelligence (Hybrid NER Pipeline)

Primary Model: BERT (Deep Learning) - High Accuracy.

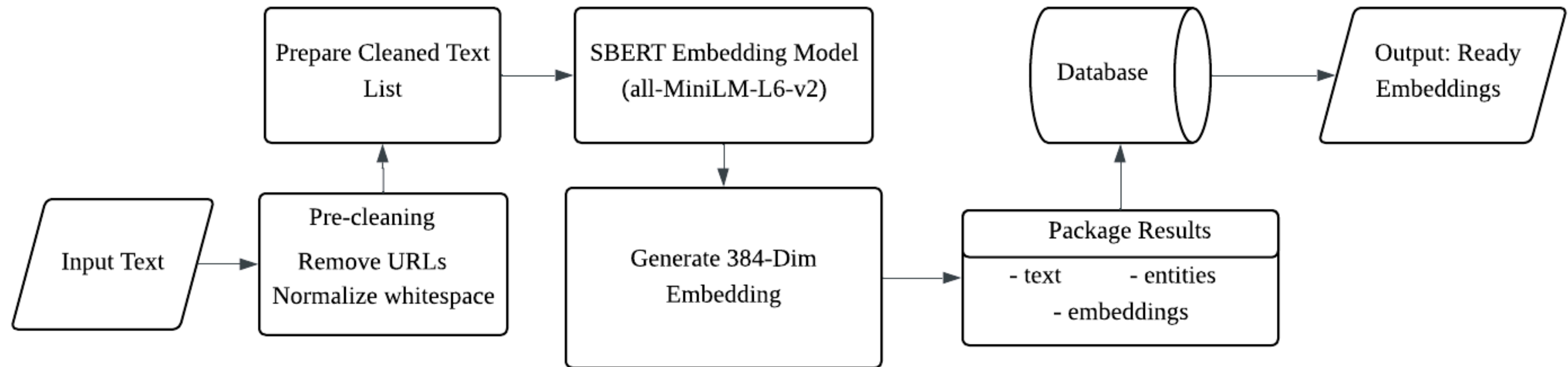
Secondary Model: spaCy (Statistical) - Fallback / Speed.



Embeddings + Vector DB (ChromaDB)

- **Embedding Generation** (Sentence-BERT)
 - Model: all-MiniLM-L6-v2 via SentenceTransformers.
 - Process: Converts raw text into a 384-dimensional vector that captures the semantic meaning of the document.
- **ChromaDB** (chroma_manager.py):
 - Manager: chroma_manager.py handles the connection to the ChromaDB server (Port 8000).
 - Storage: Saves the Document, Vector Embedding, and Metadata (JSON of entities like Names/Locations) together.

Embeddings + Vector DB (ChromaDB)



Thank You